

OTA Firmware Update Runbook

NCD LTE Sound Sensor (THS) → firmware **V1.66**

Over-the-air update · no physical access required · October 2026

What V1.66 adds: support for the Windows config app (Bluetooth pairing fix), and correct capture of the modem IMEI/ICCID in the config app. This runbook pushes it to **already-deployed Sound Sensor units** over cellular — no site visit needed.

Before you start

- The target unit must be **online and publishing** to the broker (it receives the update command on its next cycle). Offline units update whenever they next reconnect.
- This image is for the **Sound Sensor (THS)** device type **only**. Do not send it to a Tag unit.
- You need the unit's **hardware ID** (12 hex characters, e.g. `16D2DA8CFA8E`) — it's on the device label and in the config app / its MQTT topics.

Step 1 — Firmware image (already hosted)

The V1.66 Sound Sensor update image is published at this URL. Use this **exact** form (the `github.com/.../raw/` link redirects, which the modem may not follow):

```
https://raw.githubusercontent.com/bhaskar-anil429/FS_LTE/refs/heads/main/LTE_THS_V1.66.bin
```

Step 2 — Send the update command

Publish this JSON to the unit's **CMD** topic over MQTT:

Topic: `fs/topic/CMD/<HARDWARE_ID>` (e.g. `fs/topic/CMD/16D2DA8CFA8E`)

```
{"cmd":"ota_ready","url":"https://raw.githubusercontent.com/bhaskar-anil429/FS_LTE/refs/heads/main/LTE_THS_V1.66.bin"}
```

Repeat per unit, substituting each unit's hardware ID in the topic. Send the same command to every Sound Sensor you want on V1.66.

Step 3 — What happens on the device

1. On its next publish cycle the unit receives the command and downloads the image over the cellular link (~350 KB, about 1–3 minutes).
2. The bootloader (MCUboot) verifies the image, stages it, and the unit **reboots into V1.66**.
3. The new image is **confirmed** only after its first successful publish — so if the update were bad, the unit **automatically rolls back** to its previous firmware.

Step 4 — Confirm success

The unit reports its running version in telemetry — after the update its `fw_version` / `TELEMATICS` shows **V1.66**. You can also open the config app and check the firmware field. Windows pairing will work once the unit is on V1.66.

Critical — device type must match. This image is Sound Sensor firmware. Sending it to a Tag unit gives that unit the wrong application. Only send to Sound Sensor (THS) units.

Safe by design. A unit can't be bricked by this: the image is signature-checked before install, and if the new firmware can't run and publish, the bootloader reverts to the prior version on its own. The same URL is allowed up to 3 download attempts per unit.

Quick reference

Item	Value
Target	Deployed NCD LTE Sound Sensor (THS) units, online
Firmware URL	<code>https://raw.githubusercontent.com/bhaskar-anil429/FS_LTE/refs/heads/main/LTE_THS_V1.66.bin</code>
Command topic	<code>fs/topic/CMD/<hardware-id></code>
Command	<code>{"cmd":"ota_ready","url":"...LTE_THS_V1.66.bin"}</code>
Download size / time	~350 KB · ~1–3 min over cellular
Verify	Telemetry <code>fw_version</code> = V1.66
Rollback	Automatic if the new image fails to run/publish